

Recurrence Relation

$$T(n) = \begin{cases} 1 & n=1 \\ 2T(n/2) + n & n>1 \end{cases}$$

$$T(n) = 2 \boxed{T(n/2)} + n \quad \text{--- ①}$$

$$T(n) = 2T(n/2) + n$$

$$T(n/2) = \underbrace{2T(n/2^2)} + \frac{n}{2}$$

$$= 2 \left[2T(n/2^2) + \frac{n}{2} \right] + n$$

$$T(n) = 2^2 \boxed{T(n/2^2)} + n + n \quad \text{--- ②}$$

$$T(n/2^2) = 2T(n/2^3) + \frac{n}{2^2}$$

$$= 2^2 \left[2T(n/2^3) + \frac{n}{2^2} \right] + 2n$$

$$T(n) = 2^3 T(n/2^3) + 3n \quad \text{--- ③}$$

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$$= 2^2 \left[2T\left(\frac{n}{2^3}\right) + \frac{n}{2^2} \right] + 2n$$

$$T(n) = 2^3 T\left(\frac{n}{2^3}\right) + 3n \quad \text{--- ③}$$

$$T(n) = 2^k T\left(\frac{n}{2^k}\right) + kn$$

Assume

$$T\left(\frac{n}{2^k}\right) = T(1)$$

$$\therefore \frac{n}{2^k} = 1 \quad n = 2^k$$

$$k = \log n$$

$$T(n) = 2^k T(1) + kn$$

$$T(n) = n \times 1 + n \log n$$

$$\underline{O(n \log n)}$$